

Exam Choice

Student Number

2019

TRIAL
EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black pen.
- Draw diagrams using pencil.
- For questions in Section II, show all relevant working in questions involving calculations.
- NESA approved calculators may be used.

Total marks: 100

Section I – 20 marks (pages 3 – 12)

- Attempt questions 1 – 20.
- Allow about 35 minutes for this section.

Section II – 80 marks (pages 13 – 28)

- Attempt questions 21 – 38
- Allow about 2 hours and 25 minutes for this section.

Section I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A  B  C  D 

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A  B  C  D 

correct 

1. Hubble's discovery of the expansion of the Universe was made possible by his observations of:
 - (A) stellar spectral class.
 - (B) stellar luminosities.
 - (C) stellar temperatures.
 - (D) stellar spectra.

2. An observer on a platform observes a passing train as being shorter than the platform.

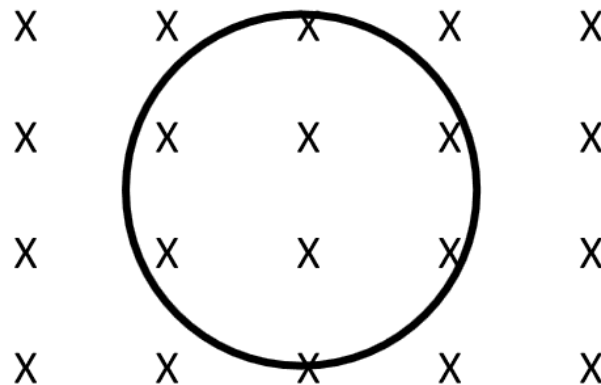
At the same time, an observer on board the train measures the platform as being shorter than the train.

This apparent contradiction is explained by:
 - (A) Einstein's statement that all inertial frames of reference are equally valid.
 - (B) The train's frame of reference is not valid as it is moving.
 - (C) Light coming to the observer from the rear of the train is travelling faster than from the front of the train.
 - (D) Neither of these frames of reference is valid.

3. One of the significant new contributions of Maxwell's equations to the understanding of electromagnetism at the time was:
 - (A) the link between magnetism and electrons.
 - (B) the discovery of radio waves.
 - (C) the prediction of an electromagnetic spectrum.
 - (D) that light was composed of photons.

4. An ideal transformer where 100% efficiency occurs is not possible as:
 - (A) eddy currents produce heat.
 - (B) the current must also change.
 - (C) transformers cannot operate using DC.
 - (D) iron cores are not magnetic.

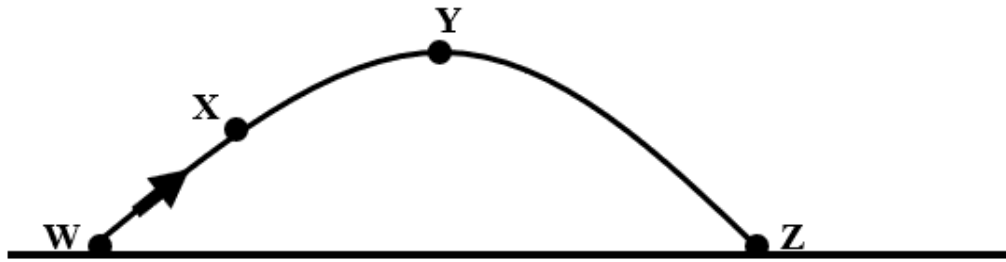
5. A circular loop of wire stands vertically within a horizontal magnetic field, as shown.



Of the following scenarios, which would produce the average greatest rate of change of magnetic flux through the same circular loop of wire?

- (A) Doubling the magnetic field intensity in 10 seconds.
 - (B) Reducing the size of the loop to zero in 10 seconds.
 - (C) Laying the loop horizontally in 10 seconds.
 - (D) Rotating the loop around a vertical axis one full turn every 10 seconds.
6. Curved train tracks are banked so that, to an occupant in the trains:
- (A) the gravitational force is matched by the required centripetal force.
 - (B) the centripetal force is equal to the normal force.
 - (C) the normal force contributes to the required centripetal force.
 - (D) the normal force aligns with the gravitational force.

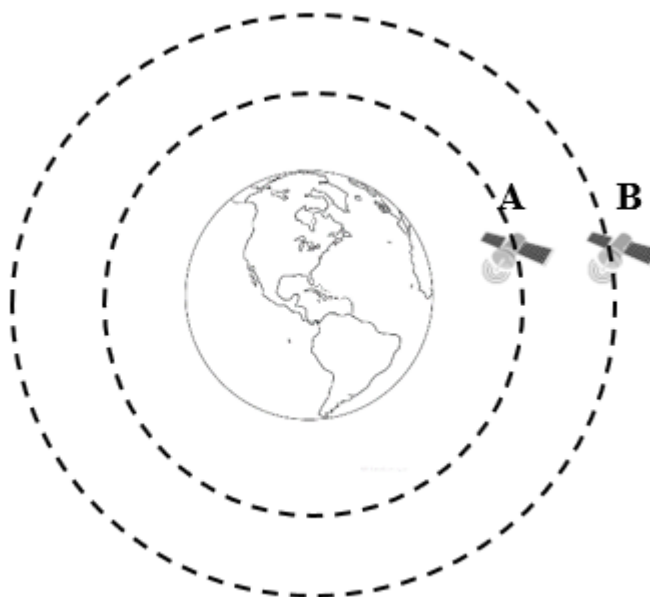
7. The flightpath of a projectile fired over level ground is shown:



The projectile has minimum speed at point:

- (A) W
 - (B) X
 - (C) Y
 - (D) Z
8. Observations of the Balmer series of hydrogen are explained by:
- (A) electrons orbiting the nucleus spiralling down and losing energy.
 - (B) electrons transitioning between energy levels.
 - (C) radiation from the nucleus.
 - (D) reflections of light photons from the electrons.

9. Two satellites, **A** and **B** are orbiting Earth at different altitudes, as shown.



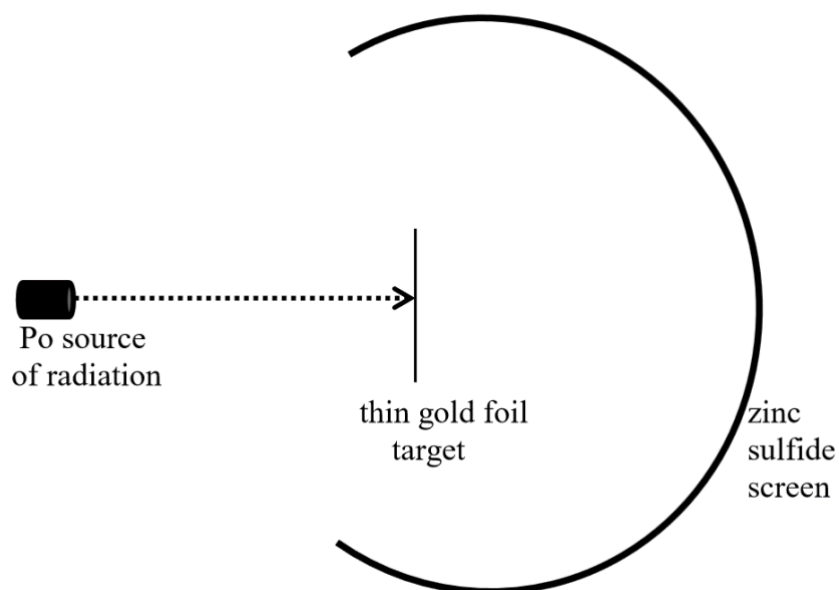
Which comparison of the satellites' orbital velocities, orbital periods and centripetal accelerations is correct?

	Orbital velocity	Orbital period	Centripetal acceleration
(A)	A is greater	B is greater	B is greater
(B)	B is greater	A is greater	A is greater
(C)	B is greater	B is greater	B is greater
(D)	A is greater	B is greater	A is greater

10. When an electron and a positron annihilate, the energy released is equal to:

- (A) $9.1 \times 10^{-31} \times 3.00 \times 10^8 \text{ J}$
 (B) $9.1 \times 10^{-31} \times (3.00 \times 10^8)^2 \text{ J}$
 (C) $(2 \times 9.1 \times 10^{-31}) \times (3.00 \times 10^8)^2 \text{ J}$
 (D) $(2 \times 9.1 \times 10^{-31} \times 3.00 \times 10^8)^2 \text{ J}$

11. An investigation was set up as shown in the diagram.



The observations made from this investigation led to:

- (A) the discovery of the nucleus.
 - (B) the discovery of the neutron.
 - (C) the discovery of cathode rays.
 - (D) the conclusion that electrons are a part of every atom.
12. Two bike riders have a race over a straight track.
- The blue bike starts from rest and accelerates uniformly at 2.0 m s^{-2} , while the red bike starts from the same position but with an initial speed of 5.0 m s^{-1} and then accelerates uniformly at 1.6 m s^{-2} .

How far from the start line are the two bikes again side by side?

- (A) 25 m
- (B) 625 m
- (C) 80 m
- (D) 32 m

13. In a particle accelerator, a proton is travelling at $0.9999c$.

It is then intentionally collided with another proton, travelling in the opposite direction but with the same speed.

What is the primary reason for colliding these two protons?

- (A) to measure their masses at such high speeds.
 - (B) to observe how they repel each other at these speeds.
 - (C) so that they collide and bounce off each other to observe if momentum is conserved or otherwise.
 - (D) so that their energy contributes to the formation of new particles.
14. The wavelength of peak intensity radiation emanating from a source with temperature T_1 is measured as λ_1 .

The temperature of the body is changed to T_2 , where $T_2 = 2T_1$.

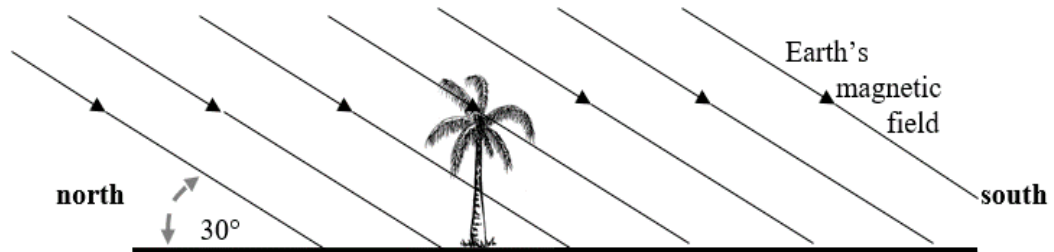
What changes in the radiation from the source would be observed?

- (A) The peak wavelength would become $2\lambda_1$.
- (B) The peak wavelength would become $\frac{\lambda_1}{2}$.
- (C) The peak wavelength would remain unchanged.
- (D) The peak wavelength would increase a small amount.

15. A student conducts an investigation into the force that acts on a section of a long straight wire.

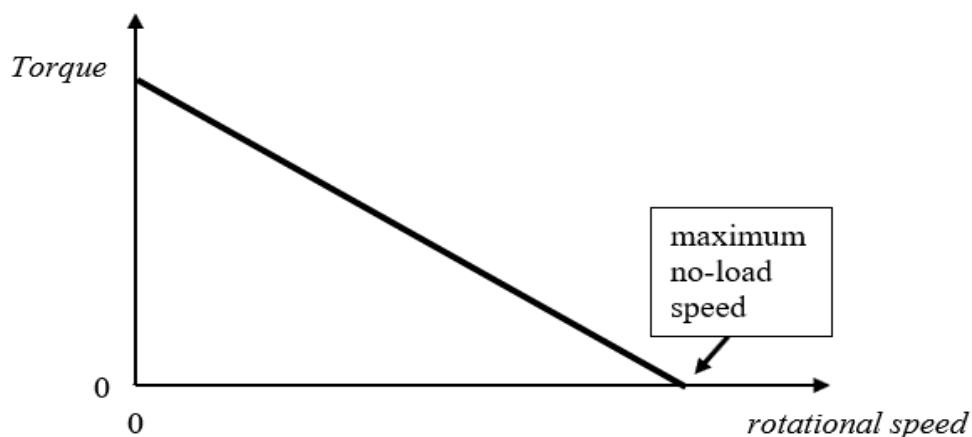
The wire carries a current and is within a magnetic field.

Where the investigation is performed, Earth's magnetic field is angled at 30° to the horizontal, as shown.



What orientation of the wire results in the maximum force on the section of wire?

- (A) vertical
 - (B) horizontal and north-south
 - (C) horizontal and east-west
 - (D) inclined so it is parallel with the magnetic field
16. A typical DC motor has a torque versus rotational speed curve shown below.



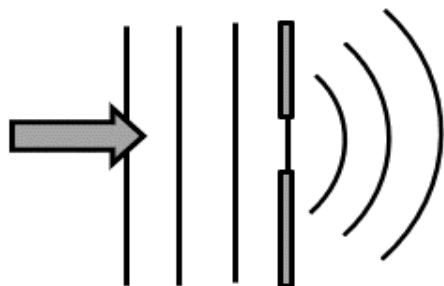
The motor having a maximum rotational speed is due to:

- (A) a slight delay in the current reversing in the coils.
- (B) back EMF.
- (C) resistance of the coils.
- (D) the coils not being sufficiently large.

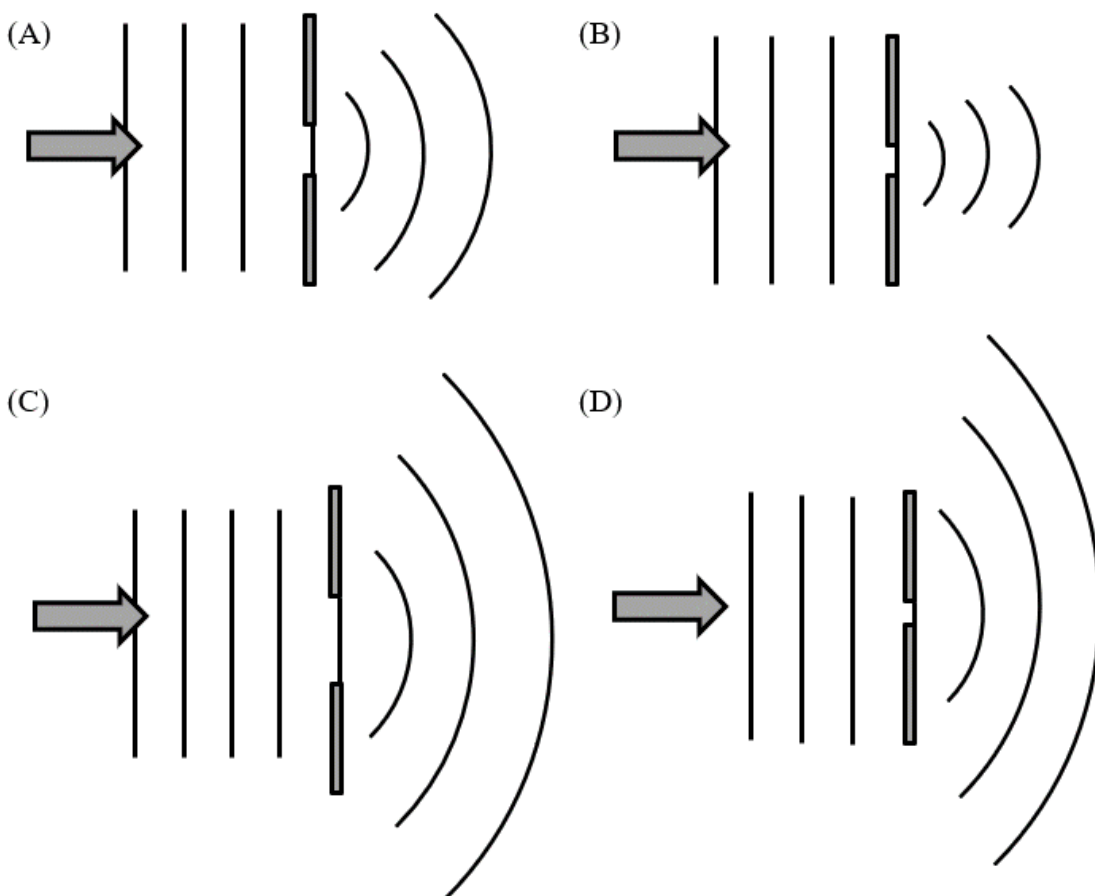
17. An investigation of diffraction of waves was performed using a ripple tank and obstacles placed in the tank.

The wavelength and slit width were varied and the results observed.

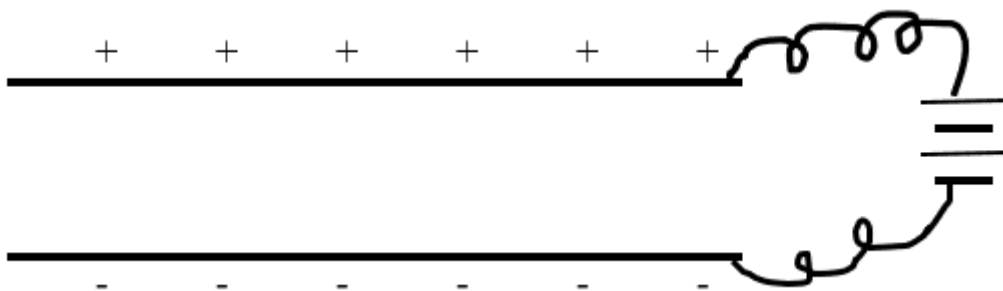
The original diffraction observations are shown.



Given the modifications made to the slit width or wavelength, which is the most likely to have been observed?



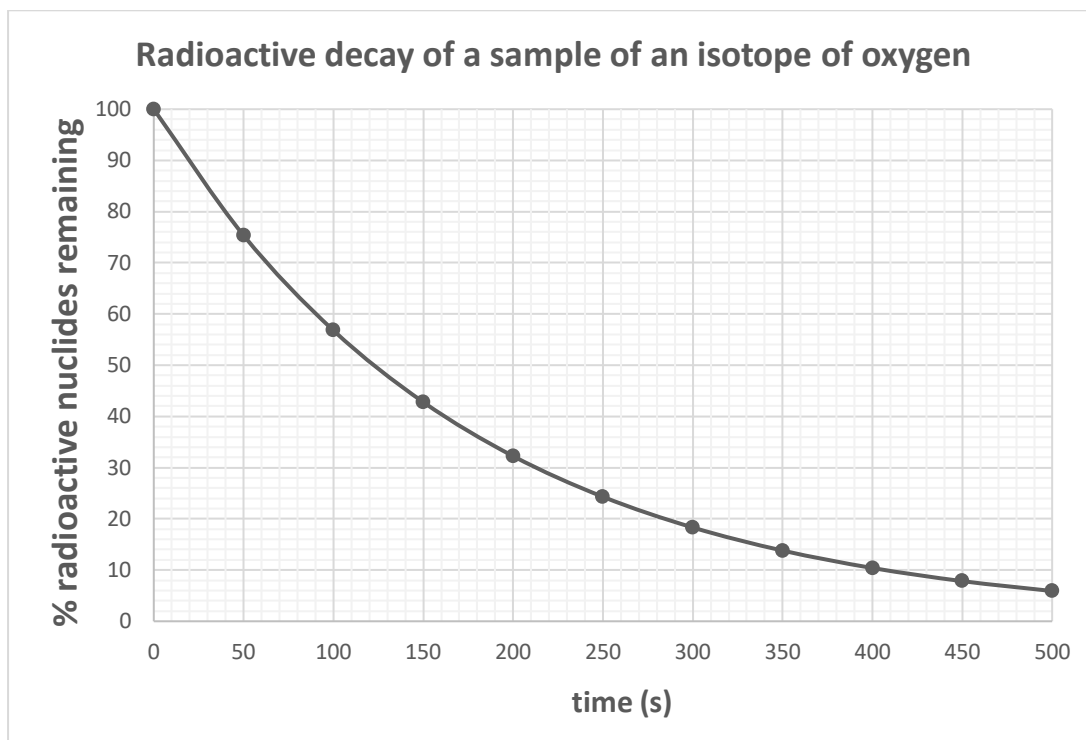
18. Two charged parallel plates have a potential difference of 35.0 V, as shown.



How far apart must these two plates be so that a point charge of $-4.0 \times 10^{-8} \text{ C}$ with a mass of $2.5 \times 10^{-6} \text{ kg}$ will be exactly suspended against gravity?

- (A) 56 cm
(B) 57 mm
(C) 17.5 m
(D) 1.8 m
19. The total potential energy of a satellite with mass $6.50 \times 10^3 \text{ kg}$ when orbiting Earth at an altitude of $2.00 \times 10^3 \text{ km}$ is closest to:
- (A) $-3.11 \times 10^{11} \text{ J}$
(B) $-3.11 \times 10^{14} \text{ J}$
(C) $-1.30 \times 10^{15} \text{ J}$
(D) $-1.55 \times 10^{11} \text{ J}$

20. The radioactivity of a small sample of an isotope of oxygen was measured over time and the results plotted on the axes shown.



What is the most likely value of the radioactive decay constant?

- (A) 125
- (B) 50
- (C) 0.005658
- (D) 0.58

2019

TRIAL
EXAMINATION

Physics

Section II Answer Booklet

80 marks

Attempt Questions 21 – 38

Allow about 2 hours and 25 minutes for this part

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which questions you are answering.

Question 21 (5 marks)

Projectile motion can be modelled by analysing the horizontal and vertical components separately.

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- (a) Show that the maximum height reached by a projectile, $h_{\max}$, is given by:

$$h_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

given: u = initial launch speed; θ = launch angle, and g = acceleration due to gravity.

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- (b) A certain projectile is always launched with a speed of 40.0 m s⁻¹.

The projectile must land on a target which is exactly 140.0 m away horizontally from the launch position.

The projectile is launched at an angle of 30°. Find whether the projectile lands on, falls short of, or travels further than the target.

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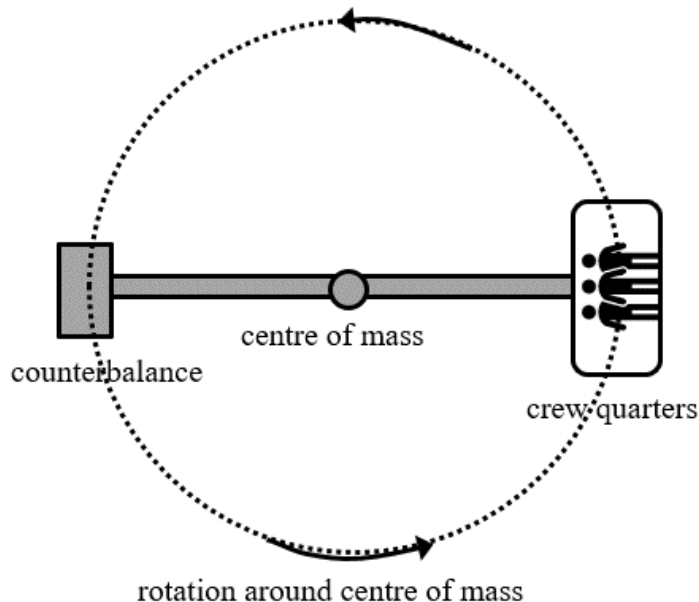
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Question 22 (4 marks)

One solution to the problem of extended periods of time in a weightless environment is to design a spacecraft which is rotating. The rotation will give rise to an “artificial gravity”.



Quantitatively derive a viable solution to this problem so that the spacecraft could rotate and the astronauts in the crew quarters would measure an artificial gravitational acceleration of 9.8 m s^{-2} . The period of rotation for the proposed solution cannot be less than one minute, or the astronauts will become disoriented and giddy.

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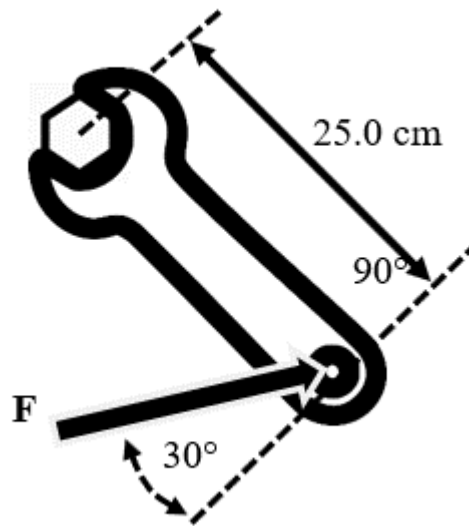
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Question 23 (2 marks)

The torque required to turn a bolt is 200 N m.

Find the force, F , required to be exerted on a spanner as shown so that the bolt will turn.

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Question 24 (9 marks)

The orbits of satellites such as GPS must be known very precisely.

With reference to the total energy of a satellite and Kepler's Laws, discuss how satellites' orbits can be predicted and maintained.

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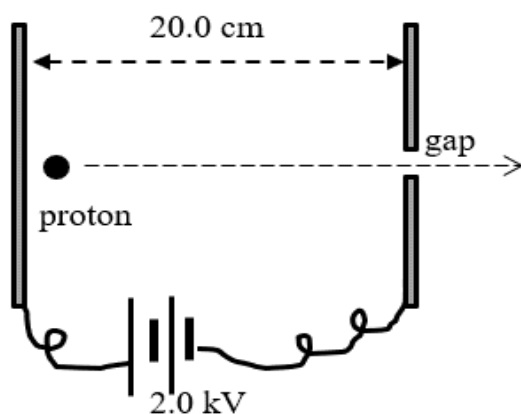
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Question 25 (4 marks)

A single proton is accelerated from rest between two charged plates which have a potential difference of 2.0 kV. The plates are 20.0 cm apart.



The proton's path leads it through the gap in the negatively charged plate.

- (a) What is the proton's speed as it passes through the gap in the negatively charged plate?

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- (b) Explain why the distance between the two charged plates has no influence on the proton's final speed.

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Question 26 (5 marks)

Transformers are designed to be as efficient as possible.

Identify the sources of energy loss and justify the design features of a transformer that improve efficiency. Include references to the underlying physics principles which allow transformers to operate.

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Marks

Question 27 (4 marks)

Discuss the relationship between Newton's Third Law of Motion and the SI definition of the ampere.

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Question 28 (4 marks)

Magnetic braking has advantages over conventional friction braking yet the conservation of energy applies to both.

Describe how magnetic braking works and how the conservation of energy applies.

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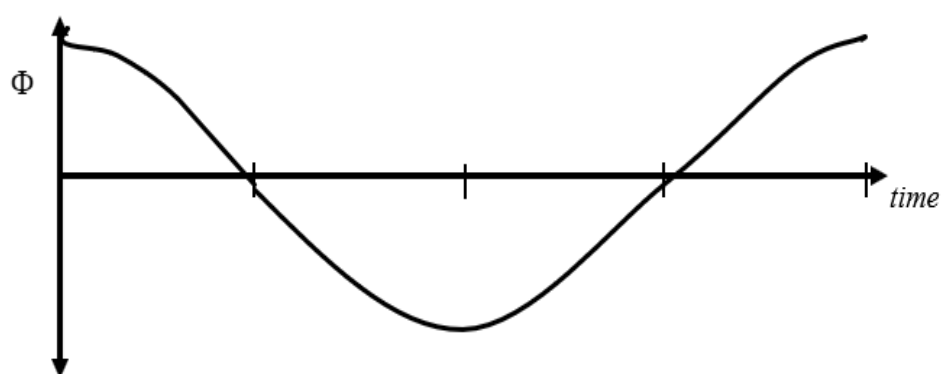
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Question 29 (3 marks)

The graph below shows the magnetic flux, Φ , against time for a rectangular loop of wire rotating within a magnetic field.



- (a) On the same axes, sketch and label the shape of the induced EMF versus time. 1
- (b) Describe *one* way in which the induced EMF graph would change if the rectangular loop was rotated faster. 2

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Question 30 (9 marks)

Analysis of the spectrum of a star can reveal information about the star, including its surface temperature, rotational and translational velocity, density and chemical composition.

Discuss how our understanding of the production of a star's spectrum and the features they contain has allowed astronomers to reveal this information.

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Question 31 (5 marks)

Outline a procedure that could be used to perform an investigation in a school laboratory to verify Malus's Law.

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Include any diagrams that may be relevant.

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Question 32 (3 marks)

In an investigation into the photoelectric effect, light of frequency 6.6×10^{14} Hz was directed onto a metal cathode. The stopping voltage was measured as 1.60V.

Find the work function of the cathode metal. Express your answer in eV.

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Question 33 (3 marks)

Analyse how atomic clocks have been able to verify time dilation as described by Einstein.

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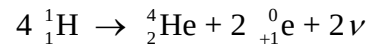
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Question 34 (4 marks)

Nucleosynthesis in main sequence stars can be summarised in the equation:



This process must occur in several steps.

Name this process and outline these steps.

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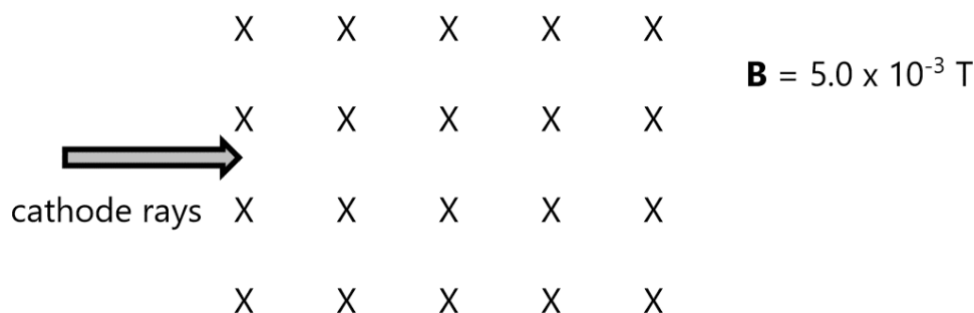
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Question 35 (4 marks)

In determining the charge-mass ratio of cathode ray particles, Thomson showed that these particles were unlike ions or alpha particles.

One part of the experiment involved firing cathode ray particles into a uniform magnetic field, as shown in the diagram.



When the cathode ray particles enter the magnetic field they are deflected in a circular path with a radius of 70.0 mm.

- (a) Explain why the shape of the path followed by the cathode ray particles is circular. 2

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- (b) Calculate the speed of these cathode ray particles. 2

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Question 36 (3 marks)

Using a typical example and an appropriate equation, show why the deBroglie wavelength of an everyday moving object is not observable.

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Question 37 (5 marks)

It has been calculated that the Sun is emitting 3.86×10^{26} J of radiative energy every second.

- (a) How much matter is being converted into energy in the Sun's core each second?

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- (b) Given that the mass of a He-4 nucleus is 6.648×10^{-27} kg, calculate how many He-4 nuclei are being produced in the core of the Sun each second.

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Marks

Question 38 (4 marks)

Outline the roles played by the four fundamental forces of nature.

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